Determinants

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March 2025

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1 Definition of a Determinant

The determinant of a square matrix A is written $|A|$ or $\det A$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \Longrightarrow \quad |A| = ad - bc, \qquad |I| = 1.$$

2 Determinants of Order Three

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

Expansion along the first row:

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Expansion along the second row:

$$|A| = -a_{21} \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} + a_{22} \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} - a_{23} \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}.$$

A determinant can be expanded along any row or column.

3 Minors and Cofactors

The minor M_{ij} is the determinant obtained by deleting row i and column j.

$$A_{ij} = (-1)^{i+j} M_{ij}.$$

The sign pattern is

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

For example,

$$M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, \qquad A_{11} = M_{11},$$

$$M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \qquad A_{12} = -M_{12}.$$

$$|A| = a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13} = \sum_j a_{ij}A_{ij}.$$

4 Area of a Triangle

For points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$,

$$\boxed{\text{Area} = \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|.}$$

Area is nonnegative; the signed determinant indicates orientation.

5 Determinant Properties

For a square matrix A of order n and scalar k,

$$|kA| = k^n|A|, \qquad |A^T| = |A|, \qquad |AB| = |A||B|.$$

If A is invertible,

$$|A^{-1}| = \frac{1}{|A|}.$$

The transpose of the cofactor matrix is the adjugate: $\operatorname{adj} A = (\text{cofactor matrix})^T$.

6 Inverse of a Matrix

$$A(\operatorname{adj} A) = (\operatorname{adj} A)A = |A|I, \qquad \boxed{A^{-1} = \frac{\operatorname{adj} A}{|A|}}, \quad |A| \neq 0.$$

For a 2×2 matrix,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

A matrix is invertible precisely when its determinant is nonzero.

7 Linear Equations Using Matrices

$$\begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3. \end{aligned}$$

$$AX = B, \qquad A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}.$$

When $|A| \neq 0$, $X = A^{-1}B = \frac{1}{|A|}(\operatorname{adj} A)B$ is the unique solution. When $|A| = 0$, inspect $(\operatorname{adj} A)B$: if nonzero there is no solution; if zero, further checking is needed to determine consistency.

8 Substitution in a Linear Equation

For $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = d$, substitute $u = 1/x$, $v = 1/y$, $w = 1/z$ to obtain $au + bv + cw = d$.